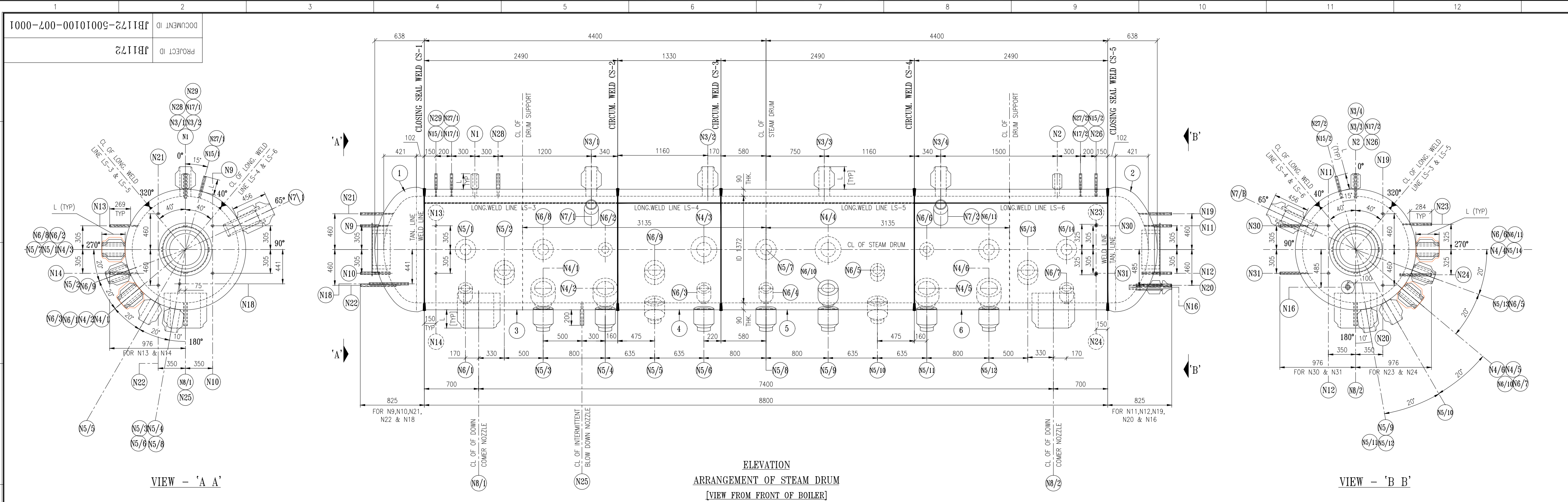
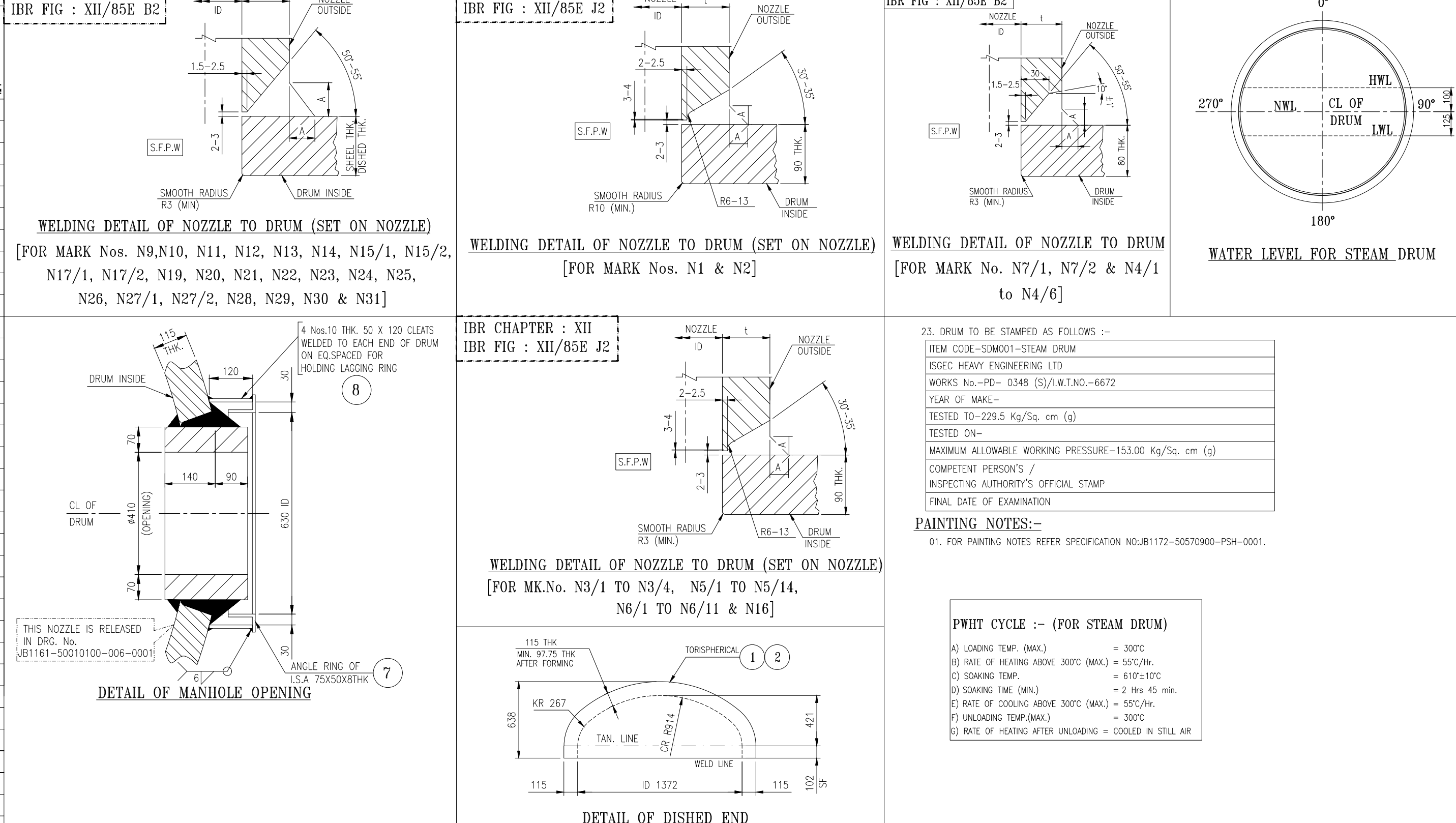


TABLE - 1									
Shos.	Nozzle Nos.	Nozzle Descriptions	Location of Nozzle	No. of Nozzle	Nozzle Size (OD) (mm)	Nozzle Thickness (mm)	Connecting Pipe Thickness (mm)	Projection from End of Drum	Projection from End of Drum
1	N1	SAFETY VALVE NOZZLE - 1	STEAM DRUM SHELL	1	140	-	44.60	-	200
2	N2	SAFETY VALVE NOZZLE - 2	STEAM DRUM SHELL	1	140	-	38.25	-	200
3	N3/1 TO N3/4	SATURATED STEAM STUB	STEAM DRUM SHELL	4	260	60	14.27	300	-
4	N4/1 TO N4/6	COMBUSTOR SIDE WALL RISER STUB	STEAM DRUM SHELL	6	315	66	18.26	300	-
5	N5/1 TO N5/14	COMBUSTOR FRONT WALL RISER STUB	STEAM DRUM SHELL	14	260	60	14.27	300	-
6	N6/1 TO N6/7	SEPARATOR SIDE, REAR & PARTITION WALL RISER STUB	STEAM DRUM SHELL	7	180	44	11.13	300	-
7	N6/8 TO N6/11	SEPARATOR FRONT WALL RISER STUB	STEAM DRUM SHELL	4	260	60	14.27	300	-
8	N7/1 & N7/2	FEED INLET NOZZLE WITH THERMAL SLEEVE	STEAM DRUM SHELL	2	327	70	REF. DRG. No.009-0001	-	-
9	N8/1 & N8/2	DOWNCOMER NOZZLE	STEAM DRUM SHELL	2	550	81	34.93	250	-
10	N9	LEVEL TRANSMITTER (STEAM)	STEAM DRUM DISHED END	1	33.4	9.09	4.55	-	350
11	N10	LEVEL TRANSMITTER (WATER)	STEAM DRUM DISHED END	1	33.4	9.09	4.55	-	350
12	N11	LEVEL TRANSMITTER (STEAM)	STEAM DRUM DISHED END	1	33.4	9.09	4.55	-	350
13	N12	LEVEL TRANSMITTER (WATER)	STEAM DRUM DISHED END	1	33.4	9.09	4.55	-	350
14	N13	LEVEL TRANSMITTER (STEAM)	STEAM DRUM SHELL	1	33.4	9.09	4.55	-	305
15	N14	LEVEL TRANSMITTER (WATER)	STEAM DRUM SHELL	1	33.4	9.09	4.55	-	305
16	N15/1 & N15/2	PRESSURE TRANSMITTER NOZZLE	STEAM DRUM SHELL	1	33.4	9.09	4.55	200	-
17	N16	CHEMICAL INJECTION NOZZLE WITH THERMAL SLEEVE	STEAM DRUM DISHED END	1	170	46.4	REF. DRG. No.009-0001	-	100
18	N17/1	PRESSURE GAUGE NOZZLE	STEAM DRUM SHELL	1	33.4	9.09	4.55	200	-
19	N17/2	REMOTE PRESSURE GAUGE NOZZLE AT FLOOR	STEAM DRUM SHELL	1	33.4	9.09	4.55	200	-
20	N18	CRD NOZZLE	STEAM DRUM DISHED END	1	33.4	9.09	4.55	-	75
21	N19	LEVEL GAUGE NOZZLE (STEAM)	STEAM DRUM DISHED END	1	33.4	9.09	4.55	-	350
22	N20	LEVEL GAUGE NOZZLE (STEAM)	STEAM DRUM DISHED END	1	33.4	9.09	4.55	-	350
23	N21	LEVEL GAUGE NOZZLE (STEAM)	STEAM DRUM DISHED END	1	33.4	9.09	4.55	-	350
24	N22	LEVEL GAUGE NOZZLE (WATER)	STEAM DRUM DISHED END	1	33.4	9.09	4.55	-	350
25	N23	HYDRA STEP NOZZLE (STEAM)	STEAM DRUM SHELL	1	33.4	9.09	4.55	-	325
26	N24	HYDRA STEP NOZZLE (WATER)	STEAM DRUM SHELL	1	33.4	9.09	4.55	-	325
27	N25	INTERMITTENT BLOW DOWN NOZZLE	STEAM DRUM SHELL	1	48.3	10.15	5.08	200	-
28	N26	NITROGEN BLANKETING	STEAM DRUM SHELL	1	33.4	9.09	4.55	200	-
29	N27/1 & N27/2	AIR VENT NOZZLE	STEAM DRUM SHELL	2	48.3	10.15	5.08	200	-
30	N28	SPARE STUB	STEAM DRUM SHELL	1	33.4	9.09	4.55	200	-
31	N29	ITC NOZZLE	STEAM DRUM SHELL	1	33.4	9.09	4.55	200	-
32	N30	LEVEL TRANSMITTER (STEAM)	STEAM DRUM SHELL	1	33.4	9.09	4.55	-	305
33	N31	LEVEL TRANSMITTER (WATER)	STEAM DRUM SHELL	1	33.4	9.09	4.55	-	305



REV.	ZONE	DESCRIPTION	REV.	ZONE	DESCRIPTION	REV.	ZONE	DESCRIPTION	REV.	ZONE	DESCRIPTION
1			2			3			4		
5			6			7			8		
9			10			11			12		
13			14			15			16		

DESIGN DATA

CODE OF CONSTRUCTION	IBR 1950 & ITS LATEST AMENDMENTS UP TO MARCH 2020 (SEE NOTE : 02)
MAXIMUM WORKING PRESSURE	153.00 Kg/Sq. cm (g)
MAXIMUM WORKING TEMPERATURE	343 °C
WORKING METAL TEMPERATURE	343 °C (OUTSIDE OF GAS PATH)
HYDRAULIC TEST PRESSURE	229.5 Kg/Sq. cm (g) (AT SHOP & AS PER IBR AT SITE)
RADIOGRAPHY OF LONG. & CIRCUM. SEAM	100%
STRESS RELIEVING	YES (REFER HEAT TREATMENT CYCLE TABLE)
EVAPORATION OF BOILER (M.C.R.)	170 TPH
TOTAL HEATING SURFACE AREA OF BOILER (EXCLUDING FORCED FLOW ELEMENT)	2855 sq.m (APPROX.)
HEATING SURFACE AREA OF FORCE FLOW SECTION (NON STEAMING)	2700 sq.m (APPROX.)
HEATING SURFACE AREA OF DRUM	NIL (OUT SIDE GAS PATH)
STEAM PRESSURE AT SUPER HEATER OUTLET	126 Kg/Sq. cm (g)
SUPER HEATER OUTLET TEMPERATURE	545 ±5 °C
WEIGHT OF DRUM (EMPTY)	39182.93 Kgs.
INSPECTION	CIB (HARYANA) (SEE NOTE : 03)

MATERIAL SPECIFICATIONS :

DESCRIPTION	SIZE	SPECIFICATION	MIN. TENSILITY (Kg./Sq.mm)	MIN.CALC. THK. (mm)	REMARKS
SHELL BELTS	AS PER MATERIAL LIST	SA 516 Gr.70	49.45	65.39	FOR STEAM DRUM
DISHED ENDS	AS PER MATERIAL LIST	SA 516 Gr.70	49.45	91.97	FOR STEAM DRUM

GENERAL NOTES :-

01. ALL DIMENSIONS ARE IN "mm" UNLESS OTHERWISE SPECIFIED.
02. THE ZERO DATE OF THIS CONTRACT IS 12.03.2020. HENCE IBR AMENDMENTS UP TO MARCH 2020 HAVE BEEN TAKEN CARE OF WHILE DESIGNING THE EQUIPMENT.
03. INSPECTION BY CIB HARYANA BECAUSE MANUFACTURING OF THIS EQUIPMENT SHALL BE DONE AT ISGEC WORKS, YAMUNA NAGAR (HARYANA)
04. ALL FILLET WELDS ARE 6mm UNLESS OTHERWISE SPECIFIED.
05. FOR DIMENSIONAL TOLERANCE REQUIREMENTS PLEASE REFER DOC. NO. GEN-50990100-05M-002 AND ITS ASSOCIATED DRAWINGS.
06. CIRCUMFERENTIAL DIMENSION (ON OUTSIDE TRAMMELS ON DRUM) STEAM DRUM - 4875.8 mm.
07. LEGEND :

H.W.L.	: DENOTES HIGH WATER LEVEL.	ID	: DENOTES INSIDE DIAMETER.
N.W.L.	: DENOTES NORMAL WATER LEVEL.	SF	: DENOTES STRAIGHT FACE.
L.W.L.	: DENOTES LOW WATER LEVEL.	CR	: DENOTES CROWN RADIUS.
CS-1&CS-5	: DENOTES CLOSING SEAM.	LS-3&LS-4	: DENOTES LONGITUDINAL SEAM.
CS-2&CS-3	: DENOTES CIRCUMFERENCE SEAM.	RS	: DENOTES RADIUS.
CS-4	: DENOTES TANGENTIAL LINE.	S.F.P.W	: SET ON FULL PENETRATION WELD

II) SHOP NOTES :

01. ALL CIRCUMFERENTIAL DIMENSIONS ARE MEASURED OUTSIDE OF DRUMS.
02. MANHOLE CIRCULAR TYP. SIZE #410 FOR DRUMS REF. DRG. No. JB1172-50010100-006-0001
03. A MAX. 4° mm FILLET WELD TO BE USED FOR WELDING OF CLEAT BRACKETS AS PER DRG.No. : 011-0001 TO 014-0001 AFTER STRESS RELIEVING.
04. THIS DRAWING SHALL BE READ IN CONJUNCTION WITH THE FOLLOWING DRGS.
 1. FOR NOZZLE & STUBS DETAILS REFER DRG.No. : JB1172-50010100-009-0001
 2. FOR EXTERNAL AND INTERNAL BAFFLES & PIPE REFER DRG.No. : 1010-0001 TO 015-0001.
05. ALL WELDING TO BE DONE BY IBR QUALIFIED WELDER USING IBR WELDING PROCEDURES
06. FOR BOILER DRUMS HAVING SHELL THICKNESS 100mm AND ABOVE FOLLOWING ADDITIONAL TESTING SHALL BE CARRIED OUT.
 - i. ULTRASONIC EXAMINATION OF ALL BUTT WELD BEFORE AND AFTER PWHT.
 - ii. MAGNETIC PARTICLE EXAMINATION OF NOZZLES/STUBS TO SHELL WELDING.
08. ALL THE PAWS WELDED ALL-ROUND ON PRESSURE PART SHALL BE PROVIDED WITH 2 Nos 17/8" NPT(F) TELL-TALE HOLES AT 180° APART ON THE PAD TO RELEASE ENTRAPPED AIR/GAS.
09. TOLERANCE FOR FORMED HEADS:-
 - i. THE INNER SURFACE OF A TORSHERICAL SHALL NOT DEVIATE OUTSIDE OF THE SPECIFIED SHAPE BY MORE THAN 1/8" OF NOMINAL INSIDE DIAMETER & INSIDE THE SPECIFIED SHAPE BY MORE THAN 3/8" OF NOMINAL INSIDE DIAMETER.
10. SUITABLE STIFFENER ARRANGEMENT SHALL BE PROVIDED AT SHOP FOR NOZZLES AT DISHED END TO AVOID DAMAGE DURING TRANSPORTATION.
11. MANHOLE DOORS ARE TO BE PLACED INSIDE THE DRUM BEFORE ASSEMBLING DRUM ENDS FOR FINAL CIRCUMFERENTIAL WELD AS PER ORIENTATION DETAIL.
12. 2mm THK. SS PLATE SHALL BE PROTECTED USING WOODEN BLANKS DURING TRANSPORTATION.
13. PRIOR TO HYDRAULIC TESTING FOLLOWING SHALL BE CARRIED OUT.
 - a) ALL WELD SPATTER, WELD STUBS, SCALE, DIRT ETC SHALL BE REMOVED FROM DRUM.
14. EQUIPMENT SHALL BE THOROUGHLY DRIED/CLEANED AFTER HYDRO TESTING.
15. HYDRO TESTING TO BE CARRIED OUT WITH GASKET IT BE THE SAME AS THE SPECIFIED SERVICE GASKET.
16. HOTTEST WATER SHALL NOT CONTAIN MORE THAN 1% SUSPENDED SOLIDS.
17. ALL INSIDE EDGES OF NOZZLES AND CONNECTIONS SHALL BE GROUND SMOOTH TO 3 MM RADIUS.
18. STRAY ARROW SHOULD BE AVOIDED. IF ANY STRAY ARROW NOTED SHALL BE DRESSED, MP & UT CHECKED.
19. FOLLOW APPROVED AND APPLICABLE WPS FOR SELECTION OF WELDING CONSUMABLES.
20. WELD SURFACES SHALL BE FREE FROM CRACKS, UNDERCUTS, UNNECESSARY GRINDING, ABRUPT RIDGES AND VALLEYS.
21. VISUAL INSPECTION OF WELD SHALL BE CARRIED OUT AND NOTED BEFORE OFFERING FOR ANY NON DESTRUCTIVE EXAMINATION.
22. HYDRO TEST WATER QUALITY SHALL BE MAINTAINED WITHIN LIMITS SPECIFIED IN DOCUMENT NO.- BLS-PPA-HYD-01-S1.

SDM001

P.NO.	PART DESCRIPTION	SPECIFICATION	SIZE	P	B215	I	39276.04	P.QTY	P.WT(Kg)	REMARK
ITEM CODE	ITEM DESCRIPTION	LTYPE	LGROUP	LQTY	LWT (Kg) (TOTAL)					REMARK

Issued to

Issued by

Issued date

For pre-planning

For manufacture in shop

For provisional manufacture in shop

For sub contracting / site work

This drawing supersedes drawing no.

Rev.

W.O. No. PD-0348

/ I.W.T. No : 6672

CONSULTANT

M/s. AVANT-GARDE SYSTEMS & CONTROLS (P) LTD.

CLIENT

M/s. RUNGTA MINES LIMITED,CHALIYAMA.

DO NOT SCALE

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FOR APPROVAL

NO

FOR INFORMATION

NO

FOR PRODUCTION

YES

FOR ERECTION

YES

RESPONSIBLE DEPT.

MECHANICAL

DRAWN BY

S.R.SRIDHARAN

CHECKED BY

R.GAMIR/R.KUMARAN

APPROVED BY

R.ARUNKUMAR

DATE

30.03.2020

TOTAL WEIGHT IN kg.

39276.04

PROJECT ID

JB1172

ELEMENT ID

50010100

I.W.T. NO.

6672

YEAR

2020

SHEET SIZE

A1

SCALE

1 : 30

SERVICE

SINGLE DRUM, WATER TUBE COMPACT CFBC BOILER

TITLE

ARRANGEMENT & DETAIL OF STEAM DRUM

DRAWING DOCUMENT ID

JB1172-50010100-007-0001

REVISION

01

CAD FILE FROM SOFTWARE : AUTOCAD